JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI VARALAKSHMI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

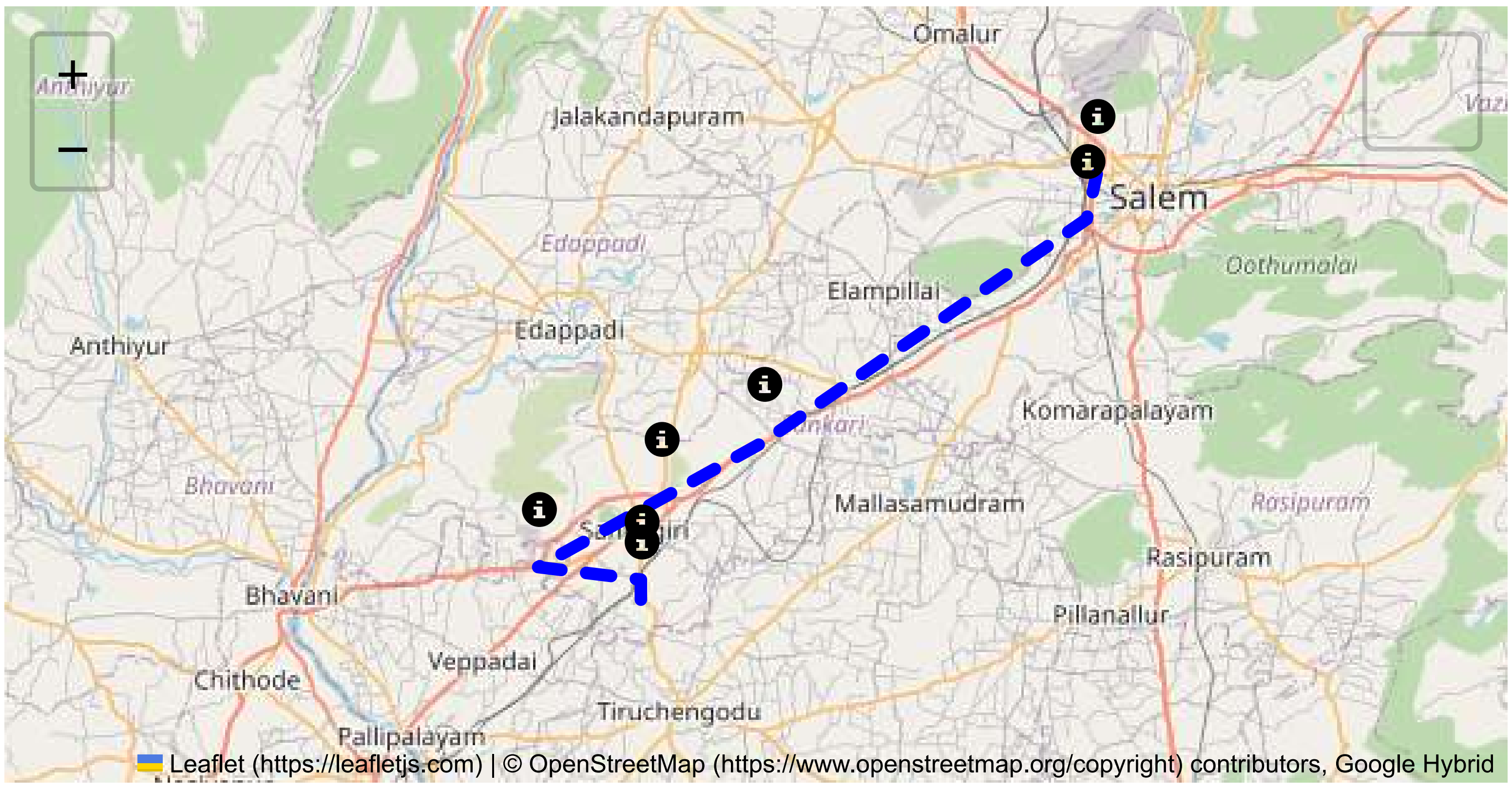
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

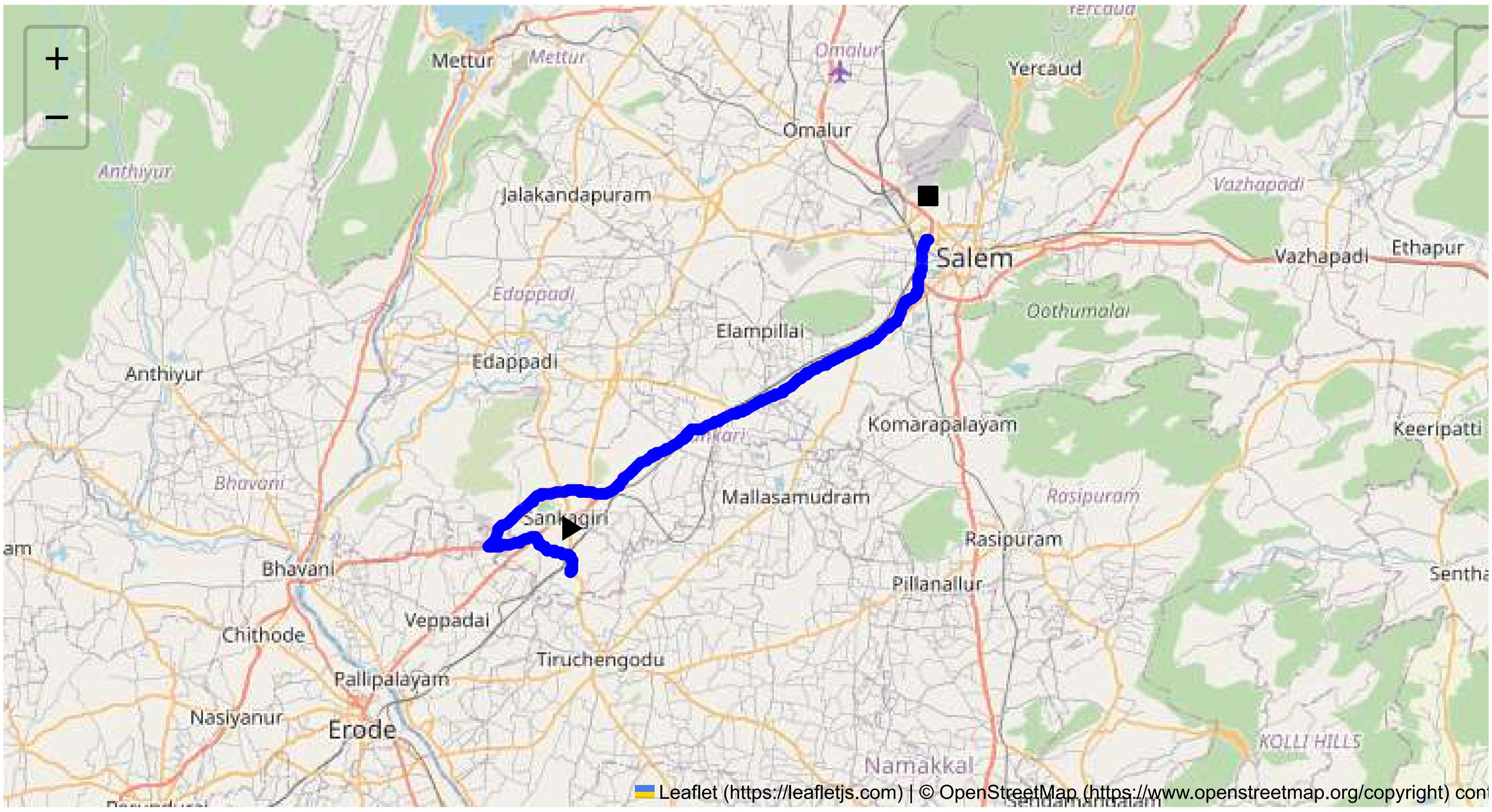
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 52.83 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.666468, 78.124451)



Welcome to the Journey Risk Management Study

Certainly! Let's analyze the route from Sangagiri to Thiruvagoundanur via several key points in Tamil Nadu for potential hazards:

1. **Overview of the Route Map:**

    - The journey covers approximately 52.83 kilometers and routes through both rural and semi-urban areas. The route uses major highways and roads connecting Sangagiri to Salem.

2. **Typical Weather Conditions:**
    - Tamil Nadu experiences tropical weather with a hot and humid climate. During the southwest monsoon (June to September), the region may experience heavy rainfall, which could lead to slippery roads and potential flooding, especially in low-lying areas.
    - The Northeast monsoon (October to December) could also bring rainfall, impacting road conditions.
3. **Traffic Patterns:**
    - Peak hours are generally in the morning (8 AM to 10 AM) and evening (5 PM to 8 PM) when traffic congestion is more likely, particularly closer to urban centers like Salem.
    - Highways and roads near market areas and junctions such as Sankari Main Road can experience slow-moving traffic, especially during weekends.
4. **Assessment of Road Quality and Infrastructure:**
    - The road quality varies; national highways are generally well-maintained, while smaller roads may have potholes or uneven surfaces.
    - The presence of construction or maintenance work might cause temporary obstructions.
5. **Suggestions for Alternative Routes in Emergencies:**
    - Depending on the exact emergency, consider alternative routes via local roads to bypass congested or hazardous areas. Connecting smaller regional roads to rejoin the primary highways might be necessary.
    - Local navigation apps often provide real-time traffic updates which can suggest alternative paths.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:**
    - Ensure compliance with Tamil Nadu's specific regulations for transporting hazardous materials. This includes proper documentation, permits, signage on vehicles, and restrictions on transport times (e.g., avoiding peak hours).
7. **Historical Incidents Involving Heavy Vehicles or Hazardous Materials:**
    - Past incidents, often due to overloading or road accidents, emphasize the need for stringent adherence to weight limits and safe driving practices. Tyre blowouts and inadequate latching are common issues.
8. **Environmental Considerations and Sensitive Areas:**
    - Maintain environmental awareness especially in rural and agricultural areas to minimize pollution.
    - Prevent contamination by ensuring robust compliance with waste management norms and accidental spill containment measures.
9. **Analysis of Communication Coverage:**
    - Most of the route is well-covered by major telecom providers. However, some rural or less developed segments might experience weaker signals. Having alternate communication means can be prudent.
10. **Estimated Emergency Response Times for Different Route Segments:**

- Urban segments such as those closer to Salem have quicker response times (15-30 minutes).
- Rural stretches might have delayed emergency services (30-60 minutes) due to distance from central facilities.

11. **Overall Summary of Risk Assessment:**

    - **Primary Risks:** Weather-related impacts during monsoon, traffic congestion especially near urban regions, and variable road quality.
    - **Moderate Risks:** Historical incidents related to overloaded heavy vehicles, slower emergency responses in rural areas, and potential communication dead zones.
    - **Mitigation Strategies:** Regularly updated navigation data, adherence to local regulations, proactive vehicle maintenance, and emergency preparedness with clear communication channels.

In summary, while transporting hazardous materials along this route, it's important to focus on weather conditions, road quality, traffic congestion, and local regulations to minimize risks effectively. Regular risk assessments and maintaining communication readiness are key to safe transit.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	U-Turn	High	11.6674963, 78.12460569999999	10 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44898, 77.87410	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.46303, 77.84945	30 KM/Hr
9	Turn	Medium	11.46318, 77.84913	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45631, 77.81668	15 KM/Hr
12	Turn	High	11.49409, 77.88004	15 KM/Hr
13	Turn	Medium	11.49419, 77.88015	30 KM/Hr
14	Turn	Medium	11.49377, 77.88573	30 KM/Hr
15	Turn	High	11.49363, 77.88590	15 KM/Hr
16	Turn	High	11.49235, 77.89561	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
18	Turn	Medium	11.61748, 78.10604	30 KM/Hr
19	Turn	Medium	11.64490, 78.12024	30 KM/Hr
20	Turn	Medium	11.64503, 78.12034	30 KM/Hr
21	Turn	High	11.66458, 78.12303	15 KM/Hr
22	Turn	High	11.66477, 78.12298	15 KM/Hr
23	Turn	High	11.66763, 78.12466	15 KM/Hr
24	Turn	High	11.66754, 78.12492	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.6674963, 78.12460569999999

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49409, 77.88004

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49377, 77.88573

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49363, 77.88590

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49235, 77.89561

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49225, 77.89560

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61748, 78.10604

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64490, 78.12024

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64503, 78.12034

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66458, 78.12303

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66477, 78.12298

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66763, 78.12466

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66754, 78.12492